Quiz 6

Graded

Student

[redacted]

Total Points

8 / 12 pts

Question 1

1a

3 / 3 pts

✔ **+ 3 pts** 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Question 2

1b

3 / 3 pts

✔ **+ 3 pts** 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Question 3

2a

1 / 3 pts

+ 3 pts 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

✔ **+ 1 pt** 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Question 4

2b

1 / 3 pts

+ 3 pts 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

✔ **+ 1 pt** 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Since we don't know the derivative of u^{x+3}, this is probably not the best choice for one of our functions. We want to break things into functions we know the derivative of (power, trig, exp, etc). This will make the chain rule easier:)

MATH 101: Quiz 6 Full name: [redacted]
Fall 2024

Show all work and clearly denote final answer. No credit will be given if there is no work shown. No calculators or outside help is permitted on this quiz.

1. Suppose that $f(4) = 4$, $g(4) = 3$, $f'(4) = -2$, and $g'(4) = 5$.

   a) Find $h'(4)$ if $h(x) = f(x)g(x)$

$$h'(4) = f'(4)\,(g(4)) + (g'(4))\,(f(4))$$

$$= (-2)(3) + (5)(4)$$

$$-6 + 20 = 14$$

   b) Find $h'(4)$ if $h(x) = \frac{f(x)}{g(x)}$

$$h'(4) = \frac{(f'(4))(g(4)) - (g'(4))(f(4))}{g(4)^2}$$

$$= \frac{(-2)(3) - (5)(4)}{9}$$

$$= \frac{-6-20}{9} = \frac{-26}{9}$$

2. Let $f(x) = (g \circ h)(x) = e^{x^2+3}$.

$g(h(x))$

a) What is $g(x)$? What is $h(x)$?

$h(x) = e^x$

$g(x) = u^{x+3}$

b) What is $f'(x)$?

$f(x) = e^{x^2+3}$

$f'(x) = x+3\, u^{x+2} (e^x)$

$= x+3 (e^{x^2+3})(e^x)$

$i = e^{x^2}$

$o = u^{x+3}$